DS4002 – Case Study
Your Name:
Date:

Can You Outsmart Fake News?

Every day, millions of people consume news online, but not all of it is real. From politics to health to global events, misinformation spreads rapidly, often disguised as legitimate reporting. As a data scientist, you are being called upon to tackle one of the most pressing challenges of the digital age: detecting fake news.

Some topics, like politics, may be filled with emotionally charged language and exaggerated claims. Others, like science or economics, may rely on more subtle distortions. This raises an important question: **are certain topics harder to classify as fake than others?**

You have been recruited by a media analytics team to investigate this problem. Using a large dataset of labeled news articles, your task is to explore how writing style, structure, and vocabulary differ between real and fake news, as well as whether those differences change depending on the topic.

To do this, you will:

- Break articles into topic-based clusters using natural language processing
- Analyze linguistic patterns within each cluster
- Build models to classify articles as real or fake
- Evaluate whether classification performance varies by topic

Your findings could help improve misinformation detection systems and provide insight into how fake news adapts across different subject areas.

The Deliverable

Your task is to produce a **data science case study report** that investigates how topic influences fake news detection. This includes:

- A structured analysis of real vs fake news characteristics
- Topic clustering using text data
- Classification models applied both globally and within topics
- A comparison of model performance across clusters
- A discussion of which topics are most difficult to classify and why

By completing this case study, you will take on the role of a data scientist working on developing tools that could help combat misinformation at scale.